Energy Conversion and Management 152 (2017) 354–365

Contents lists available at ScienceDirect

Energy Conversion and Management

journal homepage: www.elsevier.com/locate/enconman

CFD analysis of radiation impact on Stirling engine performance

R. Ben-Mansour, A. Abuelyamen, Esmail M.A. Mokheimer*

Mechanical Engineering Department, King Fahd University of Petroleum and Minerals, Dhahran 31261, Saudi Arabia

ARTICLE INFO

Keywords:
Stirling engine β-type
CFD
Radiation
Air, helium and hydrogen
Emissivity

ABSTRACT

The impact of radiative heat transfer on the performance of a β-type Stirling engine with no regenerator was investigated numerically using ANSYS fluent 14.5 software. Two radiation models were tested, which are, namely, surface-to-surface 'S2S' and discrete ordinates 'DO'. The CFD model was validated against experimental data and the validation revealed an excellent agreement with the experimental data with a maximum deviation of 7%. On the other hand, the results revealed that ignoring radiative heat transfer results in underestimation of the power output by about 13% compared with the pertinent experimental data used for validation. Moreover, the simulation results of another geometry revealed an underestimation of the output power of about 18% for the case of ignored radiation effects compared with the results of the case with the radiation model considered. The experimentally validated model has been used to assess the engine performance using three different gases (Air, He and H_2) as working fluids. For each case, the performance is evaluated with three values of material emissivity (0.6, 0.8 and 1), and compared with the case where radiative heat transfer is ignored. It was found that DO radiation model is less expensive computationally and it is more accurate than S2S radiation model. It was shown that the radiative heat energy represents 9%–12% of the total supplied energy in the case where air gas is used depending on the surface emissivity. This percentage is reduced to 7%–9% for the case of He gas. While it didn't exceed 2% in the case of H_2 gas. Also, the results show that the power output changes proportionally with the surface emissivity in the case of air and He gases. While in the case of H_2 gas, the impact of surface emissivity can be ignored for all emissivity range under study.

1. Introduction

Stirling engine started to attract researchers due to its remarkable characteristics. It is classified as an external combustion engine. Therefore, it can use different heat sources such as solar energy, waste heat or heat energy coming from the combustion of fuels. The most implementation of Stirling engine is to generate electric power, especially by utilizing Sun energy as a heat source. Many researches were conducted in the field of designing and optimizing Stirling engine and its accessory from the technical and economical point of views [1–6]. National Aeronautics and Space Administration (NASA) also pay attention to develop a Stirling power conversion technology for potential future space missions. Where they claimed that the higher conversion efficiency of the Stirling cycle could reduce the PuO_2 fuel by fourfold [7].

In general, the heat transfers through conduction, convection, and radiation. The radiative heat transfer contributes noticeably to the energy transfer when there is high-temperature level such as that associated with the combustion of gases in engines, furnaces, high-temperature heat exchangers, and during the explosion of chemicals. Moreover, radiative heat transfer can be important even at low temperature levels and it plays a major role in explaining different phenomena. For instance, ice could be formed from water in plastic cover at nighttime. Considering convection heat transfer only would fail to explain this phenomenon without radiative heat transfer, where heat loss occurs by emitting energy to the cold heat sink of the night sky. In radiative heat transfer, the energy transfers by means of electromagnetic waves, hence, it does not require a medium like convective and conductive heat transfer.

There are five methods used to design and predict the performance of Stirling engine; zero-order analysis, first-order analysis, second-order analysis, third-order analysis (or nodal design method) and fourth order analysis (or CFD simulation) [8,9]. The term order here is a kind of classification according to the complexity of the model as the order increases.

Sage code is a 1D model which is used to simulate and optimize a resonant system of a spring-mass-damper and it can also be used to model the performance of Stirling engine [10].

Dyson et al. [11] performed one of the fastest CFD simulation, which is for an axisymmetric free-piston Stirling engine. In a

* Corresponding author.
E-mail addresses: rmansour@kfupm.edu.sa (R. Ben-Mansour), ahmedsalih45@yahoo.com (A. Abuelyamen), esmailm@kfupm.edu.sa (E.M.A. Mokheimer).

http://dx.doi.org/10.1016/j.enconman.2017.09.056
Received 26 June 2017; Received in revised form 19 August 2017; Accepted 19 September 2017
Available online 30 September 2017
0196-8904/ © 2017 Elsevier Ltd. All rights reserved.

Nomenclature

a	absorption coefficient
c_p	specific heat at constant pressure (kJ·kg^{-1}·K^{-1})
g	Gravity acceleration (m·s^{-2})
k	thermal conductivity (W·m^{-1}·K^{-1})
l_1, l_2, l_3, l_4	lengths of the linkages of the rhombic drive mechanism (m)
l_d	height of displacer (m)
n	refractive index
p	pressure (Pa or bar)
q	heat flux (W·m^{-2})
r_1, r_2	radius of displacer and cylinder (m)
$\vec{r}$	position vector
s	path length
$\vec{s}$	direction vector
$\vec{s}'$	scattering direction vector
t	time (s)
u, v	velocity components in x- and r-direction (m·s^{-1})
$\tilde{u}, \tilde{v}$	relative velocity in x- and r-direction (m·s^{-1})
u_c, v_c	moving frame velocity components in x- and r-direction (m·s^{-1})
u_p, u_d	piston and displacer velocities (m·s^{-1})
x, r	cylindrical coordinates in x- and r-direction
A	surface area (m^2)
F_{kj}	view factor between surfaces k and j
G	regenerative channel gap (m)
I	radiation intensity
L	distance between two gears (m)
L_{dt}	length from the linkage l_4 to top surface of the displacer (m)
L_{pt}	height from the linkage l_1 to top surface of the piston (m)
L_t	height of the engine (m)
Q_{in}, Q_{out}	heat absorb and rejected from the Stirling engine (W)
Q_i	total heat transfer during step i (W)
Q_{rad}	total radiative heat transfer (W)
R	gas constant (J·kg^{-1}·K^{-1})
R_d	offset distance from the crank to the center of gear (m)
T, T_C, T_H	temperature, cold-end temperature, and hot-end temperature (K)
V	volume (m^3)
W_{out}	power output (W)

Greek symbols

ε	emissivity
η	thermal efficiency (%)
θ	crank angle (degree or radian)
θ_k & θ_j	angle between the surface normal and a ray between the two differential areas
μ	viscosity (kg·m^{-1}·s^{-1})
ρ	density (kg·m^{-3})
σ	Stefan-Boltzmann constant (5.669e-8 W·m^{-2}·K^{-4})
σ_s	scattering coefficient
ω	rotational speed (rad·s^{-1})
Φ	phase function
Ω	solid angle

Subscripts

f	fluid
rad	radiative heat

comparison between their results and those obtained by Sage code, it turns out that axisymmetric CFD simulation is more accurate than Sage code results.

Chen et al. [12] conducted a CFD simulation for a 3D model of gamma-type Stirling engine. For the sake of simplification, they studied the heat transfer characteristics with the assumption of having no regenerator. They showed that there is dissimilarity between CFD results and those obtained by the second-order analysis, particularly in heat transfer rates. Accordingly, they clarified that the one-dimensional models are not accurate due to the non-uniform temperature distribution, which is the main assumption of the 1D model. On the other hand, Salazar and Chen [13] highlighted the poor assumption of constant heat transfer coefficient in the second -order analysis due to non-uniform temperature distribution. Also, their results showed that for a beta-type Stirling engine with no regenerator, the thermal efficiency could reach up to 14%. Abuelyamen et al. [14] performed a parametric study for a similar geometry to that reported in [13]. Their results showed that the optimum charge pressure varies according to the gas type used as the working fluid to run the engine. Moreover, they concluded that the power output could be increased by shifting the temperature-end-limits to a higher level, but in return, the efficiency goes down. Moreover, a complex geometry of an alpha-type Stirling engine was examined numerically by Mahkamov and Ingham [15,3]. They concluded that the power output predicted by second-order analysis is twice the value calculated by CFD analysis. Moreover, a prior manufactured gamma-type Stirling engine was built based on a first-order model analysis. Unexpectedly, the real generated power was found to be very small compared with the design value. Hence, Mahkamov [16] investigated the performance of this engine numerically. He found that the hydraulic losses in the regenerator and the dead volume are the main factors that reduce the power output. Thus, Mahkamov changed the geometry from gamma-type to an alpha-type in an attempt to reduce the dead volume, and therefore increase the power output. Lin et al. [17] simulated a 3-D model of free-piston Stirling engine charged with He gas. Their results showed high heat supply value compared with the experimental value.

A moving regenerator (with porous medium) in a beta-type Stirling engine was examined by Chen et al. [18]. The results revealed that high porosity value leads to high pressure losses, however, it enhances the thermal efficiency and the power output. More studies on Stirling engine regenerator have been conducted by Costa et al. [19]. They concluded that the regenerator of stacked shape has superior performance compared with the wound woven wire matrices regenerator. Furthermore, they compared the performance of a stack woven regenerator and wound woven wire matrices to be used in Stirling engine [20]. They considered laminar flow for cases where *Re* is small and turbulent model for cases where *Re* is large. They also found that for *Re* range between 20 and 400, the stack woven matrices reflected a higher value of *Nu* number (by about 20%) compared to that of the wound woven matrices. Additionally, their results reflected a minor influence of volumetric porosity on *Nu* number, but they related this to the small range of volumetric porosity under study (0.6–0.68). Their model was validated against an empirical relationship of Tanaka et al. [21] and Gedeon and Wood's work [22] in terms of heat transfer coefficient. Another work of Costa et al. [23] showed that the wire mesh diameter and *Re* number have an impact on the friction factor of stack woven wire. Hence, they developed a pressure drop correlation that includes the two parameters (wire mesh diameter and *Re*). Costa et al. [24] carried out an experimental and numerical work on Stirling engine including a wound woven wire matrix regenerator. They developed a correlation between *Re* number and the friction factor for laminar flow ($5 < Re < 50$) and porosity range of (0.52–0.72). Further work on Stirling engine which includes a compact porous-sheets heat exchanger was performed by Li et al. [25]. It is an alpha-type where He gas was adopted as working fluid and Ross-mechanism was selected to govern

the motion of the pistons. They summarized that the porous sheet regenerator produces less entropy than conventional wire mesh regenerator, and therefore, it has high thermal efficiency and power output compared with the conventional wire mesh regenerator.

Ibrahim [26] tried to conduct a CFD simulation of an experimental test rig (developed by Jiang and Simon [27]). The real geometry has radial oriented heater slots, they tried to approximate it as a 2-D axisymmetric problem by using same hydraulic diameter. The results revealed a good matching between the CFD results and the experimental results but a large distance from the heater. Also, they concluded that large pressure loss occurs in the regenerator part when the piston has high velocity.

It is clear from the above literature that most of the analysis regarding Stirling engine ignores the internal radiative heat transfer [11–13,18,28,29]. Those who considered the radiation heat transfer are focusing only on calculating external radiation coming from the Sun or an external combustion heater to use it as a heating source [1–6,30–34], but they do not deal with the radiation heat inside the Stirling engine itself.

The aim of this work is to explore the impact of radiative heat transfer inside a β-type Stirling engine on its performance. This study is bridging the gap of ignoring radiative heat transfer in the analysis of Stirling engine. Also, the current study helps in better understanding of the heat transfer characteristic of Stirling engine, because it discusses the phenomena of heat transfer inside Stirling engine from the perspective of both radiative and convective modes. The study is accomplished by examining three values of surface emissivity (0.6, 0.8 and 1). Additionally, the impact of radiative heat transfer is investigated for three gasses (Air, He and H_2). Also, a comparison between two radiation models is carried out in this work in order to identify the appropriate one for Stirling engine. The importance of this study appears as a result of the absence of such research in the field of Stirling engine as well as a result of the need for a more accurate model for Stirling engine.

Table 1
Dimensions of the β-type Stirling engine in (mm).

r_1	20
r_2	20.5
G	0.5
$l_1 = l_2 = l_3 = l_4$	18
L_t	158
L	42
L_{pt}	50.93
L_{dt}	163.74
l_d	79.46
R_d	3.5

2. Problem formulation

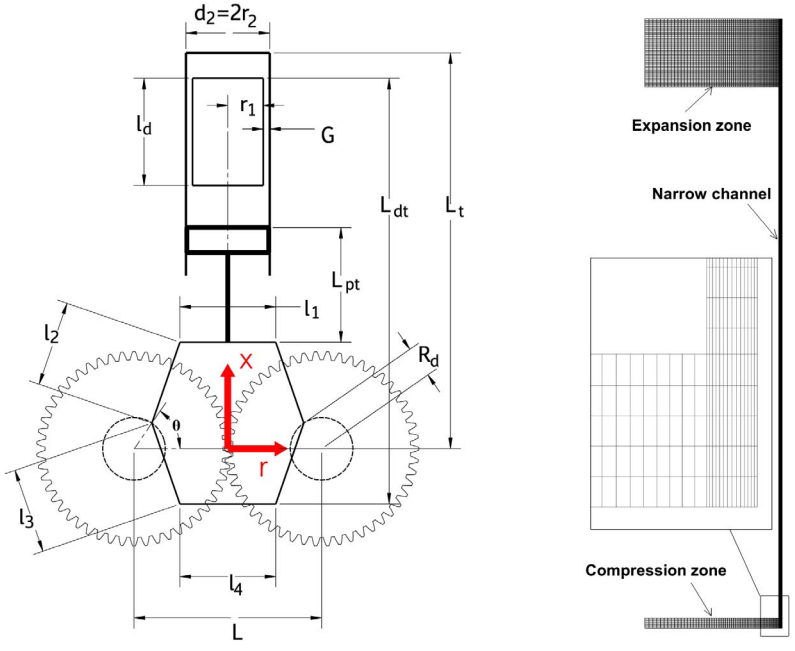
2.1. Geometry description

In order to evaluate the impact of radiative heat transfer on the performance of Stirling engines, a β-type Stirling engine driven by a rhombic mechanism is adopted as an example. It consists of three zones; compression zone, expansion zone, and narrow zone, which connects the expansion and compression zones as shown in Fig. 1. In general, this narrow channel zone is filled with regenerator materials, however, here it is assumed to have no regenerator. From the same Figure, it easy to notice that the problem could be solved as axis-symmetric geometry. So that, only half of the domain is simulated. The dimensions of the simulated geometry are reported hereunder in Table 1.

Three different gases are utilized in this work, which are namely; air, He and H_2. They are treated as an ideal gas. Additionally, the thermal properties of these gases are assumed to be dependent on the gas temperature.

The thermal boundary conditions are limited between hot temperature (T_H) and cold temperature (T_C) of 800 K and 300 K, respectively. Accordingly, the external cylinder walls are exposed to a variable temperature, which is a function of axial coordinate (x) x-coordinates:

$$T(x) = \begin{cases} T_C(\mathrm{K}), & \text{if } x \leqslant 0.08 \text{ m} \\ T_C + \frac{x-0.08}{0.158-0.08}(T_H - T_C)(\mathrm{K}), & \text{if } x > 0.08 \text{ m} \end{cases} \quad (1)$$



(a) Engine configuration

(b) Computational domain

Fig. 1. Stirling engine β-type geometry configuration.

For all cases under study, the Stirling engine is set to rotate at a speed of 1800 rpm with atmospheric charge pressure.

The cylinder walls have a thickness of 1 mm and it is made of steel. It has density of 7840 kg/m^3, thermal conductivity of 43 W·m^{-1}·K^{-1} and specific heat of 450 J·kg^{-1}·K^{-1}. Since the main purpose of this study is to reflect the impact of radiative heat transfer on the performance of Stirling engine, three material emissivity values are tested (0.6, 0.8 and 1).

2.2. Governing equations

The transport equations describing the flow field and heat transfer phenomena inside Stirling engine are embodied mathematically by transient axisymmetric compressible Never-Stokes equations, conservation of energy equation, conservation of mass and ideal gas equations. Those equations can be written as:

The continuity:

$$\frac{\partial \rho_f}{\partial t} + \frac{\partial}{\partial x}(\rho_f \tilde{u}) + \frac{1}{r}\frac{\partial}{\partial r}(\rho_f r \tilde{v}) = 0 \tag{2}$$

The momentum:

$$\begin{aligned} &\tfrac{\partial}{\partial t}(\rho_f u) + \tfrac{\partial}{\partial x}(\rho_f \tilde{u} u) + \tfrac{1}{r}\tfrac{\partial}{\partial r}(\rho_f r \tilde{v} u) = \rho_f g - \tfrac{\partial p}{\partial x} + \mu\tfrac{\partial}{\partial x}\left(\tfrac{\partial u}{\partial x}\right) + \tfrac{u}{r}\tfrac{\partial}{\partial r}\left(r\tfrac{\partial u}{\partial r}\right) \\ &\qquad + \tfrac{\mu}{3}\tfrac{\partial}{\partial x}(\nabla . V) \\ &\tfrac{\partial}{\partial t}(\rho_f v) + \tfrac{\partial}{\partial x}(\rho_f \tilde{u} v) + \tfrac{1}{r}\tfrac{\partial}{\partial r}(\rho_f r \tilde{v} v) = -\tfrac{\partial p}{\partial r} + \mu\tfrac{\partial}{\partial x}\left(\tfrac{\partial v}{\partial x}\right) + \tfrac{u}{r}\tfrac{\partial}{\partial r}\left(r\tfrac{\partial v}{\partial r}\right) \\ &\qquad + \tfrac{\mu}{3}\tfrac{\partial}{\partial r}(\nabla . V) - \tfrac{\mu v}{r^2} \end{aligned} \tag{3}$$

The energy:

$$\begin{aligned} \frac{\partial}{\partial t}(\rho_f T_f) + \frac{\partial}{\partial x}(\rho_f \tilde{u} T_f) + \frac{1}{r}\frac{\partial}{\partial r}(\rho_f r \tilde{v} T_f) &= -\frac{1}{c_{pf}}\left[\frac{\partial p}{\partial t} + \nabla . (pV) - p(\nabla . V)\right] \\ &+ \frac{k_f}{c_{pf}}\left[\frac{\partial}{\partial x}\left(\frac{\partial T_f}{\partial x}\right) + \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T_f}{\partial r}\right)\right] \\ &+ \frac{\mu}{c_{pf}}\left\{2\left[\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial r}\right)^2 + \left(\frac{v}{r}\right)^2\right]\right. \\ &\left. + \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial r}\right)^2 - \frac{2}{3}(\nabla . V)^2\right\} \end{aligned} \tag{4}$$

Ideal gas equation:

$$pV = \rho_f R T_f \tag{5}$$

where $\tilde{u} = u - u_c, \tilde{v} = v - v_c$ are the relative velocity components between the fluid and moving frames with velocities of u_c and v_c in the axial direction (x) and the radial direction (r), respectively.Two types of radiation modeling were examined in this work. The first one is a surface-to-surface (S2S) model, which account for the radiation exchange in an enclosure of gray-diffuse surfaces. In S2S model any absorption, emission or scattering of radiation by the medium can be passed over. Therefore, the heat flux leaving a given surface (k) is composed of emitted and reflected energy as follows:

$$q_{out,rad,k} = \varepsilon_k \sigma T_k^4 + \rho_k \sum_{j=1}^{N} F_{kj} q_{out,rad,j} \tag{6}$$

where $q_{out,k}$ is the energy flux leaving surface (k), $q_{out,j}$ is the heat flux from the surroundings, ε_k is the emissivity, σ is the Stefan-Boltzmann constant and F_{kj} is the view factor which is the fraction of energy leaving surface (k) that is incident on surface (j). It is calculated as:

$$F_{kj} = \frac{1}{A_k}\int_{A_k}\int_{A_j}\frac{\cos\theta_k \cos\theta_j}{\pi r^2}\delta_{kj} dA_k dA_j \tag{7}$$

where $\delta_{kj} = 1$ if dA_j is visible to dA_k and zero otherwise.

The second radiation model that is used in this study is Discrete Ordinates (DO) radiation model. The DO model solve the radiative heat transfer equation by taking into account the effect of absorption, emission, and scattering. However, in the present work, the coefficients of absorption, emission and scattering are set to zero because the working fluid is not radiation participating medium.

The general form of the radiative transfer equation (RTE) for an absorbing, emitting, and scattering medium is:

$$\frac{dI(\vec{r},\vec{s})}{ds} + (a+\sigma_s) I(\vec{r},\vec{s}) = an^2\frac{\sigma T^4}{\pi} + \frac{\sigma_s}{4\pi}\int_0^{4\pi} I(\vec{r},\vec{s}')\Phi(\vec{s}\cdot\vec{s}')d\Omega' \tag{8}$$

For the gray diffuse radiation, the net radiative heat flux leaving the surface is given by:

$$q_{out,rad} = (1-\varepsilon_w) q_{in,rad} + n^2 \varepsilon_w \sigma T_w^4 \tag{9}$$

where q_{in} is the incident radiative heat flux:

$$q_{in} = \int_{\vec{s}\cdot\vec{n}>0} I_{in}\,\vec{s}\cdot\vec{n}\, d\Omega \tag{10}$$

In general, the DO model is more computationally expensive than S2S model because it transforms RTE into a transport equation for radiation intensity, then it is solved as any transport equation since there are direction $\vec{s}$. However, when S2S model used with CFD dynamic mesh problems, it becomes more computationally expense than DO model because the view factor needs to be calculated at each time step due to the changes in the geometry. Again, for the DO model, the working fluid (medium) inside Stirling engine does not participate in radiative energy, so that the scattering, emitting, and absorbing coefficients would be zero. Accordingly, only radiation between surfaces would be solved.

Initially, the gas is at 300 K and cylinder walls temperature is given by Eq. (1), which is not a function of time. The cylinder-top and the cylinder-bottom were kept at T_H and T_C all the time, respectively. Additionally, displacer is assumed to be insulated such that no heat transfer will go through it. Moreover, the movement of the piston and displacer are governed by a rhombic motion as shown in Fig. 1. The velocity of this kind of motion is represented by:

$$\begin{aligned} u_p(t) &= R_d\omega\left\{\cos\theta - \sin\theta\left[l_2^2 - \left(\tfrac{L}{2} - \tfrac{l_1}{2} - R_d\cos\theta\right)^2\right]^{-0.5}\left[\tfrac{L}{2} - \tfrac{l_1}{2} - R_d\cos\theta\right]\right\} \\ u_d(t) &= R_d\omega\left\{\cos\theta + \sin\theta\left[l_2^2 - \left(\tfrac{L}{2} - \tfrac{l_4}{2} - R_d\cos\theta\right)^2\right]^{-0.5}\left[\tfrac{L}{2} - \tfrac{l_4}{2} - R_d\cos\theta\right]\right\} \end{aligned} \tag{11}$$

The left gear in Fig. 1 rotates in a clockwise direction, as a result, a negative value of the rotation speed (ω) was used in above equations. The motion starts from a crank angle of zero degree.

On the other hand, the cylinder walls of Stirling engine are exposed to a conduction heat transfer in the radial direction and its energy equation for this case is given by Fourier's law:

$$q = -k\frac{\partial T}{\partial n} \tag{12}$$

Since the partial Eqs. (2)–(4) have been solved numerically, the supplied and rejected heat transfer were calculated as following:

$$\begin{aligned} Q_{in} &= \tfrac{\omega}{2\pi}\sum_{i=1}^{N} Q_i . \Delta t \quad if \quad q > 0 \\ Q_{out} &= \tfrac{\omega}{2\pi}\sum_{i=1}^{N} Q_i . \Delta t \quad if \quad q < 0 \end{aligned} \tag{13}$$

where Δt is time step. Q_{in}, $Q_{out,}$ and Q_i are the total heat added, total heat rejected and total heat transfer, respectively, during the step (i). Q_i is calculated across the outer surface of the domain. The software gives the value of Q_i in (W) or (J/s). It is then multiplied by Δt to obtain the amount of energy transferred during the step (i). Therefore, the summation value for all number of steps (N) of one cycle gives the total

energy produced during that cycle in a unit of (J/cycle). Accordingly, it multiplies by (ω/2π), which represents the number of revolutions in each one second. While the power output is calculated numerically from:

$$W_{out} = \frac{\omega}{2\pi} \sum_{i=1}^{N} 0.5(p_{i-1} + p_i).(V_i - V_{i-1}) \quad (14)$$

where p is the average pressure at the specified zone, and V is the volume of that zone. The term in the summation (in Eq. (14)) is representing the average area under the P-V diagram. It is the average pressure of time step (i) and (i-1) multiply by volume differences occurring in the same steps (i and i-1). The summation of these value for a total number of steps for one cycle (N) gives the work output per cycle (J/cycle). Then, the power output is obtained by multiplying the summation term by (ω/2π).

3. Solution methodology

In order to solve the continuity, energy and momentum Eqs. (2)–(4) in addition to the view factor and radiative heat transfer Eqs. (7) and (8), the general purpose code of ANSYS fluent 14.5 was used. The program uses the finite volume method to discretize these differential Eqs. (2)–(4) and (8). The second order upwind finite difference scheme was adopted for continuity, momentum and energy equations. To couple the pressure and velocity, a coupling scheme was utilized because it shows a relatively fast convergence compared with a SIMPLE scheme. The SIMPLE scheme uses a pressure-based segregated algorithm in which the momentum equation and the pressure correction equation are solved separately. This semi-implicit solution method results in slow convergence [35].

Furthermore, variable thermal properties were implemented in the software. The ideal gas model, which is represented by Eq. (5) was used to calculate the density while the kinetic theory was used to identify the heat capacity and the viscosity. In addition, the effect of viscous heating dissipation (the last term in Eq. (4)) was tested and it turns out that there is a little bit difference in the results, thus, it can be negligible. Moreover, the gravity effect was tested as well and it turns out to be of no impact on the engine performance. Therefore, no viscous heating nor gravity effects were included in the present study.

In addition, laminar flow is assumed because the maximum *Re* based on the hydraulic diameter of the narrow channel does not exceed the laminar range for all of the cases under study. However, in some cases, it becomes in the transitional range (2000–4000) but it does not convert to fully turbulent. The temperature profile described by Eq. (1) and the movement of the piston and the displacer (Eq. (11)) are implemented in the software through a user-defined function (UDF). The conduction heat transfer through the external walls, which is described by Eq. (12) has not been included in the computational domain. Nevertheless, the wall properties and its thickness were defined in the wall boundary condition such that the software will solve the conduction heat transfer without the need to be drawn in computational geometry.

4. Mesh independence and validation studies

4.1. Mesh and time independence study

Before starting the simulation of Stirling engine, mesh and time independence study were carried out. Four different mesh size were tested. They have a number of cells of 2740, 6869, 9600 and 15020 cells. All meshes studied included dense mesh in the narrow space consisting of 10, 14, 16 and 20 intervals, respectively. The aspect ratio for all meshes under study was kept under 11.8. Moreover, each one of these mesh sizes was tested with three step-number per cycle of 360, 480 and 720 step/cycle which is corresponding to time-step intervals of 9.26×10^{-05}, 6.94×10^{-05} and 4.63×10^{-05} s, respectively. It was found that the mesh size of 6869 cells with 480 step/cycle (i.e. 6.94×10^{-05} s) achieves the mesh and time-step independent solution with a deviation of 0.6% in generated power compared with the finest mesh size with the smallest time step.

In addition, it is worth mentioning here that for the simulated Stirling engine, the domain is scaled up by 10^{10} then it returns back to the original size in Fluent software by scaling down by 10^{-10} in order to compensate 32-bit precision available in grid generation especially when there is fraction length as recommended by [11].

4.2. Radiation parameters independence study

The accuracy of calculating the view factor in S2S radiation model depends on different parameters. Also, the solution of the discretized RTE, when DO radiation model is used, depends on some parameters. Accordingly, a sensitivity analysis for these parameters is required to ensure solution independence of these parameters. In S2S radiation model, the three parameters that control the accuracy of view factor are the resolution, the subdivisions, and the normalized separation distance (NSD). The NSD is the ratio of the minimum distance between the two faces to effective face diameter. If the NSD value is less than 5, the hemicube method calculates the view factor more accurately than ray tracing method. In the proposed mesh, there are some faces have NSD less than 5, therefore, hemicube method is adopted to calculate the view factor with an NSD value of 5. The resolution and subdivisions parameters are tested for values of (10 and 5), (15 and 8), and (20 and 10). On the other hand, in the DO model, the RTE solution should be independent of angular discretization parameters (number of divisions and pixels). For this reason, values of (2 and 1), (4 and 2), and (8 and 4) are tested for divisions and pixels values, respectively.

Inspecting from the net heat transfer profile, we found out that S2S radiation model parameters achieve accurate view factors by 15 and 8 as values for resolution and subdivisions, respectively. These values produce a maximum error of 0.65% in the net heat transfer value. The error, here, is calculated under the assumption that the right solution is the solution, which has the highest number of divisions and pixel values (case of 20 division and 10 pixels). While the angular discretization values for DO model becomes independent of divisions and pixel numbers by values of 4 and 2, respectively. The maximum error in the net heat transfer profile, for this case, is 0.24%.

4.3. Validation against an experimental study

Although there are many experimental works conducted on Stirling engine [26,36–40], there was very few experimental data in the literature that fully describes the prototype geometry and gives a complete definition of all operating and boundary conditions. These information are required as inputs for the CFD simulations. In this regard, the experimental work of Aksoy and Cinar [39] provides an adequate description of the information required for the CFD simulation. Aksoy and Cinar [39] examine the performance of β-type Stirling engine which is govern by a rhombic mechanism. Table 2 summarized the specifications of their data that are used to validate the current CFD

Table 2
Engine specifications that are applied in the validation case [39].

T_H (K)	773
T_C (K)	300
Working fluid	Air
Rotational speed (rpm)	408
r_1 (mm)	43
r_2 (mm)	42.25
l_d (mm)	155
$l_1 = l_2 = l_3 = l_4$ (mm)	66
R_d (mm)	l_1/2.6666

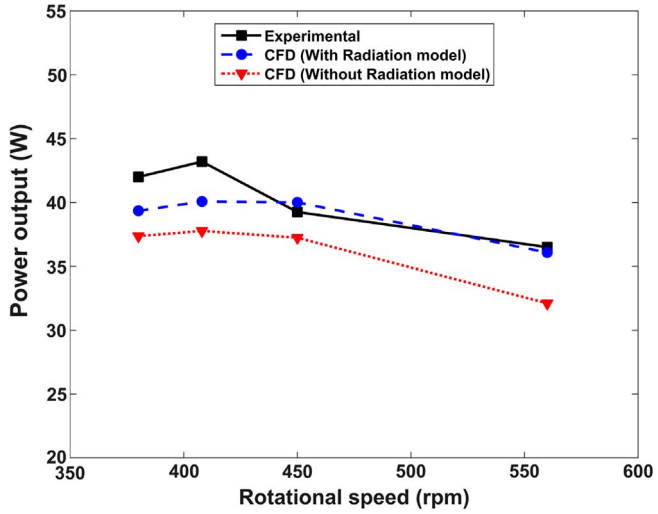


Fig. 2. Comparison of experimental data of [39] with the proposed CFD models.

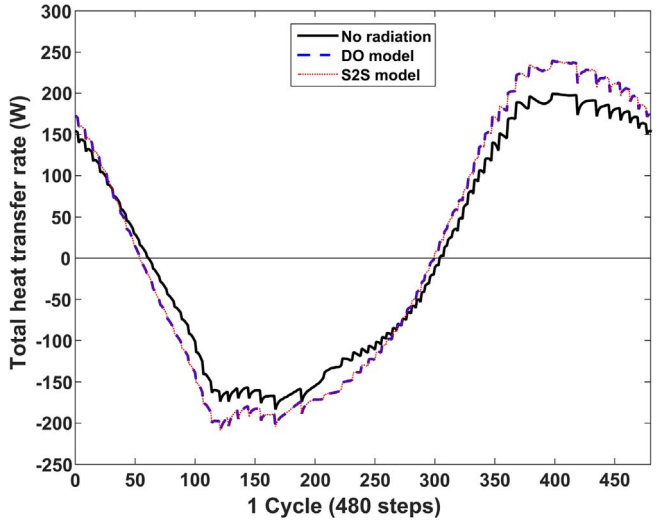


Fig. 3. Heat transfer profile obtained by different radiation models.

model. Nevertheless, neither the heating area nor the cooling area are declared in [39]. Therefore, they are approximated from the sketch provided in their report.

In the CFD simulation, the two flow conditions (laminar and turbulent) are examined, since the flow condition was not recognized to be laminar or turbulent based on the data in Table 2. The results show that the *Re* number exceeds the laminar range in some part of the cycle, therefore, a turbulent model of k-ε is implemented in the validation case. Also, the surface roughness of the cylinder and the displacer might play a role in converting the flow to turbulent condition. Moreover, the radiation model is also included in the CFD simulation, because the radiative heat transfer occurs inherently in each experimental work.

In order to ensure that the model is viable, further investigation on the power output is conducted and compared with the experimental data of power output reported by Aksoy and Cinar's [39]. They

Table 4
Stirling engine performance using air gas with different emissivity values.

	e = 1.0	e = 0.8	e = 0.6	No radiation
Q_{in} (W)	116.0	112.5	109.2	100.4
Q_{out} (W)	−106.4	−103.3	−100.4	−92.6
Q_{net} (W)	9.6	9.1	8.8	7.9
Efficiency (%)	8.3%	8.1%	8.0%	7.8%

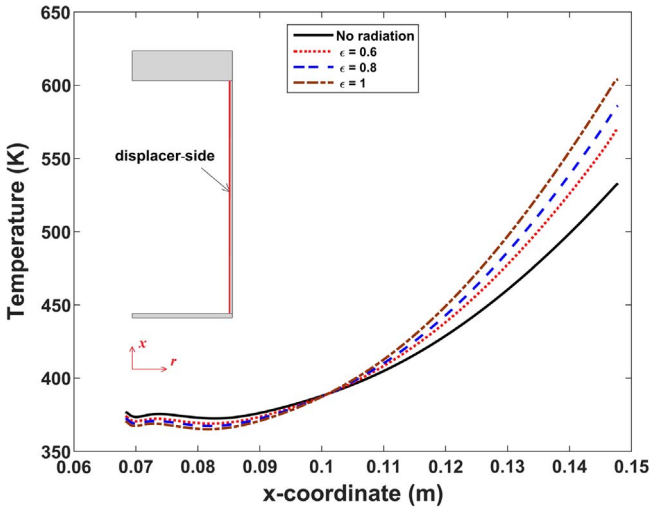


Fig. 4. Temperature profile at the displacer-side (at step 4800) using air gas.

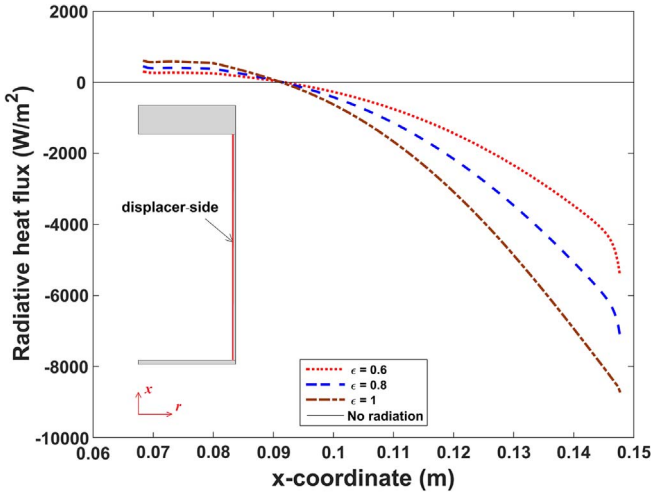


Fig. 5. Radiative heat flux on the displacer-side (at step 4800) using air gas.

developed a theoretical model to predict the power output using heat transfer coefficient raged from 200 to 400 W/m^2. Their results were represented for a heat transfer coefficient raged from 200 to 400 W/m^2, while the experimental data indicated that only 104 W/m^2 as heat transfer coefficient is required to have a good matching between their theoretical model and the experimental results. Due to the aforementioned reason, the comparison of the current work with the work of

Table 3
Stirling engine performance with different radiation models.

	Q_{in} (W)	Q_{out} (W)	Q_{net} (W)	$\int P\,dV$ (W)	Efficiency (%)	Error in work
No radiation	100.42	−92.56	7.86	8.16	7.8%	3.8%
DO model	116.04	−106.42	9.61	9.90	8.3%	3.0%
S2S model	115.93	−106.41	9.52	9.90	8.2%	4.0%

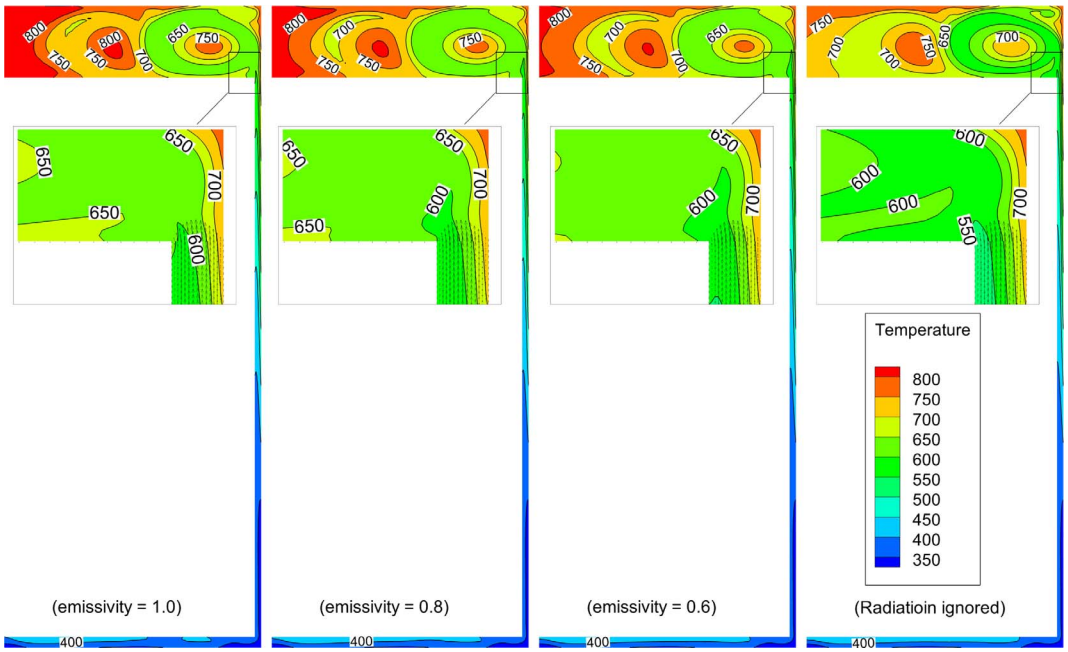


Fig. 6. Temperature contours (at step 4800) using air gas.

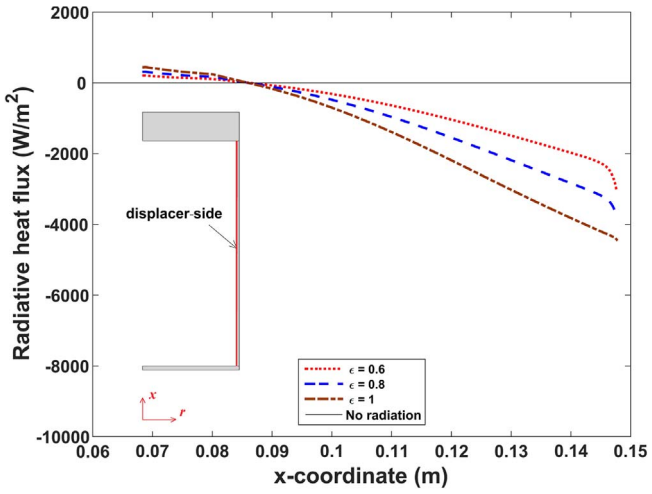


Fig. 7. Radiative heat flux on the displacer-side (at step 4800) using He gas.

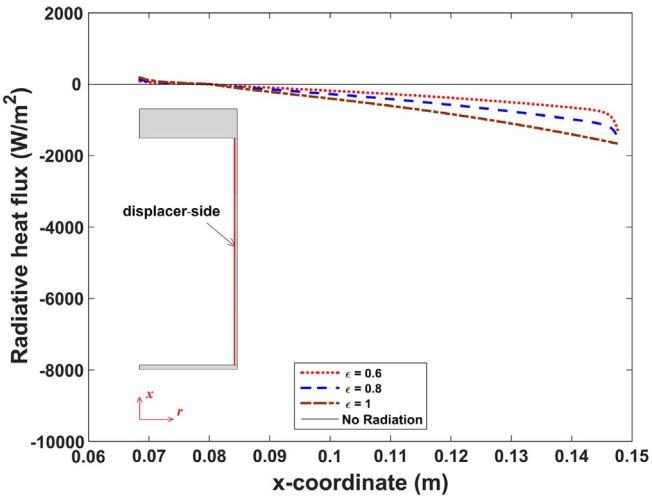


Fig. 9. Radiative heat flux on the displacer-side (at step 4800) using H_2 gas.

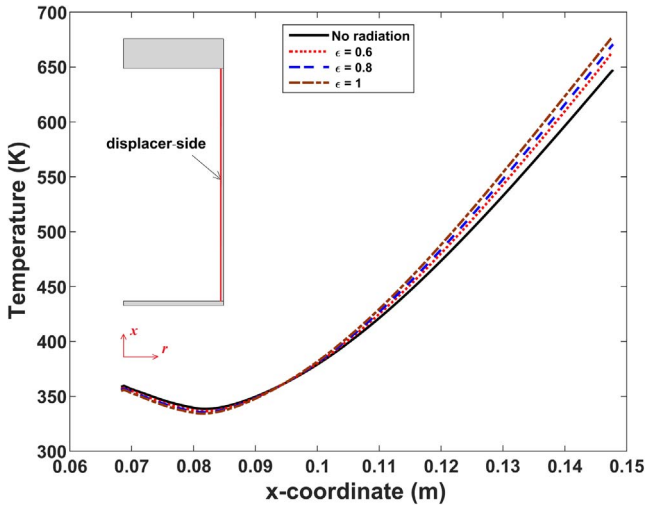


Fig. 8. Temperature profile on the displacer-side (at step 4800) using He gas.

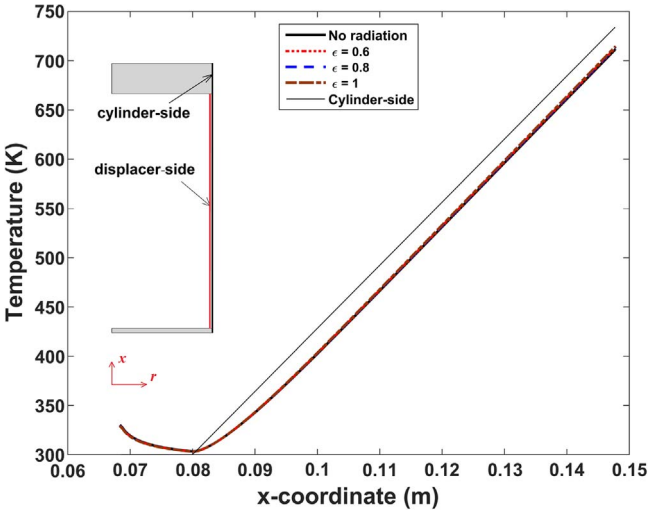


Fig. 10. Temperature profile on the displacer-side (at step 4800) using H_2 gas.

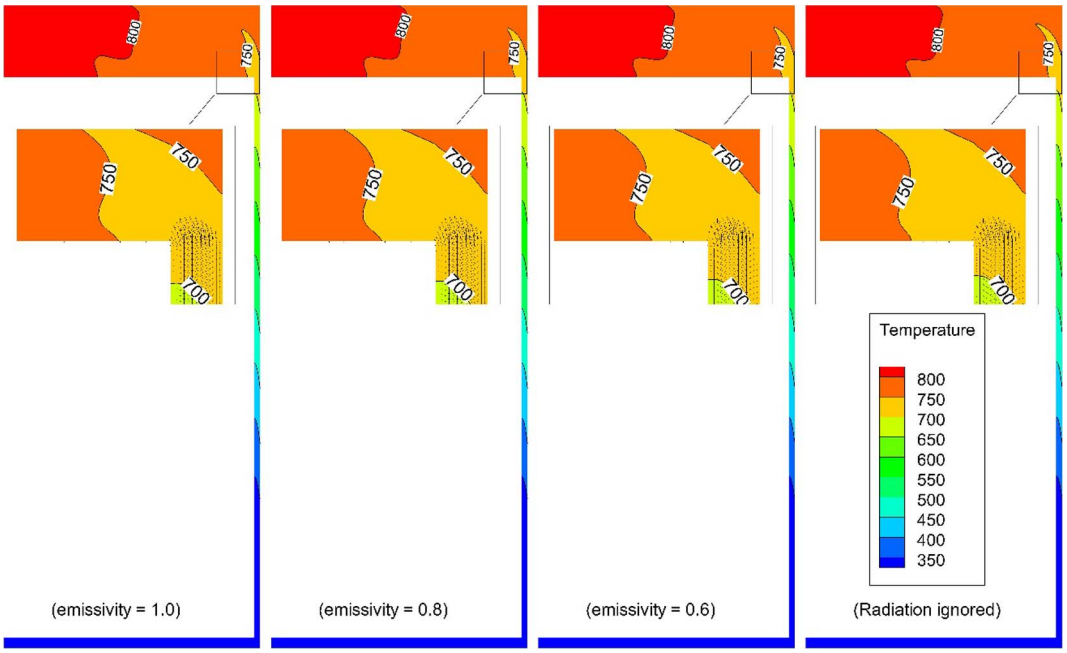


Fig. 11. Temperature contours (at step 4800) using H_2 gas.

Table 5
Stirling engine performance using He and H_2 gases with different emissivity.

	He		H_2	
	Qnet (W)	Efficiency (%)	Qnet (W)	Efficiency (%)
$\varepsilon = 1.0$	14.8	14.3%	17.6	10.2%
$\varepsilon = 0.8$	14.7	14.5%	17.6	10.3%
$\varepsilon = 0.6$	14.6	14.6%	17.6	10.4%
No radiation	14.1	14.9%	17.4	10.4%

[39] is limited only to their experimental data. Fig. 2 compares the power output calculated by the present models of CFD with the experimental data at different rotation speed (380, 408, 450 and 560 rpm). It is clear that the proposed 'DO radiation model' is being able to capture a close value of the power output at each rotational speed similarly as in the experimental data. The results of the CFD simulation with the radiation considered and modeled using the DO radiation model and the CFD simulation ignoring the radiation have the same trend as that of the experimental results. The maximum power output occurs at 408 rpm. Then it decreases as the rotational speed increases. The DO radiation model predicted a power output of 40.07 W which is deviated from the pertinent experimental value of 43.2 W with a deviation of only 7%. On the other hand, the S2S radiation model recorded a power output of 38.89 W with a deviation 8.1% from the pertinent experimentally measured value. Moreover, when the radiation heat transfer is omitted from the CFD model, the predicted power is estimated to be 37.77 W with a deviation of about 13% from the pertinent experimental value. The validation with the experimental results revealed that ignoring the radiation model results in a large deviation between the experimental results and the CFD simulation results, which emphasizes the importance of radiation model.

5. Results and discussion

Having the developed CFD code validated, many simulations have been conducted in order to explore the effect of radiation on the performance of the beta-type Stirling engine driven by rhombic mechanism. Fig. 3 depicts the heat transfer profiles obtained by S2S and DO models compared with the case of ignored radiative heat transfer. The positive value in Fig. 3 is the absorbed heat, while the negative value represents the rejected heat. It is easy to notice that the two radiation models curves are almost identical. However, the S2S model slightly underestimates the supplied energy. The S2S model simulation revealed that the engine absorbs about 115.93 W, while in the DO model simulation revealed that around 116.04 W of heat is transferred into the engine. Also, it can be noticed from Fig. 3 that ignoring the radiative heat transfer in the energy equation results in a small value of the total absorbed and rejected heat compared to the cases where radiative heat transfer is considered. Consequently, the output power would be less than the output power obtained when radiative heat transfer is taken into consideration. In addition to these heat transfer comparisons, the power output and efficiency comparisons are demonstrated in Table 3. In all cases, there are differences between the net heat transfer (Q_{net}) and the power output calculated from PV-diagram. For instance, the DO model reflects a net heat transfer value of 9.61 W, while the calculated PV power is 9.90 W (3.0% deviation). The deviation obtained by DO model is lower than the deviation obtained when no radiation heat transfer is considered (3.8% deviation). Thus, one can conclude that the DO model is able to diminish the deviation between net heat transfer and the power output calculated by PV-diagram. In contrast, the S2S model showed a large deviation between Q_{net} and PV-diagram power (4%). Moreover, in S2S radiation model, the view factors are required to be calculated at each time step, which is highly expensive computationally. Accordingly, for the sake of high accuracy and low computational expense, the DO model radiative heat transfer is adopted for further analysis in the current work.

Furthermore, taking the values obtained by the DO model as the most accurate values (since it was the closest to the experimental measurements), one can conclude that ignoring radiation energy yields a deviation of 13% in gained and rejected heat transfer. While the useful power output and efficiency are deviated by 18%. These high values of errors clarify the importance of considering the radiative heat transfer in the performance analysis of Stirling engine.

Having the importance of taking the radiative heat transfer into consideration while analyzing the performance of Stirling engine, the study is, then, extended to examine different values of surface emissivity (0.6, 0.8, and 1). The surface emissivity value plays a major role in absorbing, emitting, and reflecting heat flux through radiation mode. The results show that as the radiation heat transfer is included in the

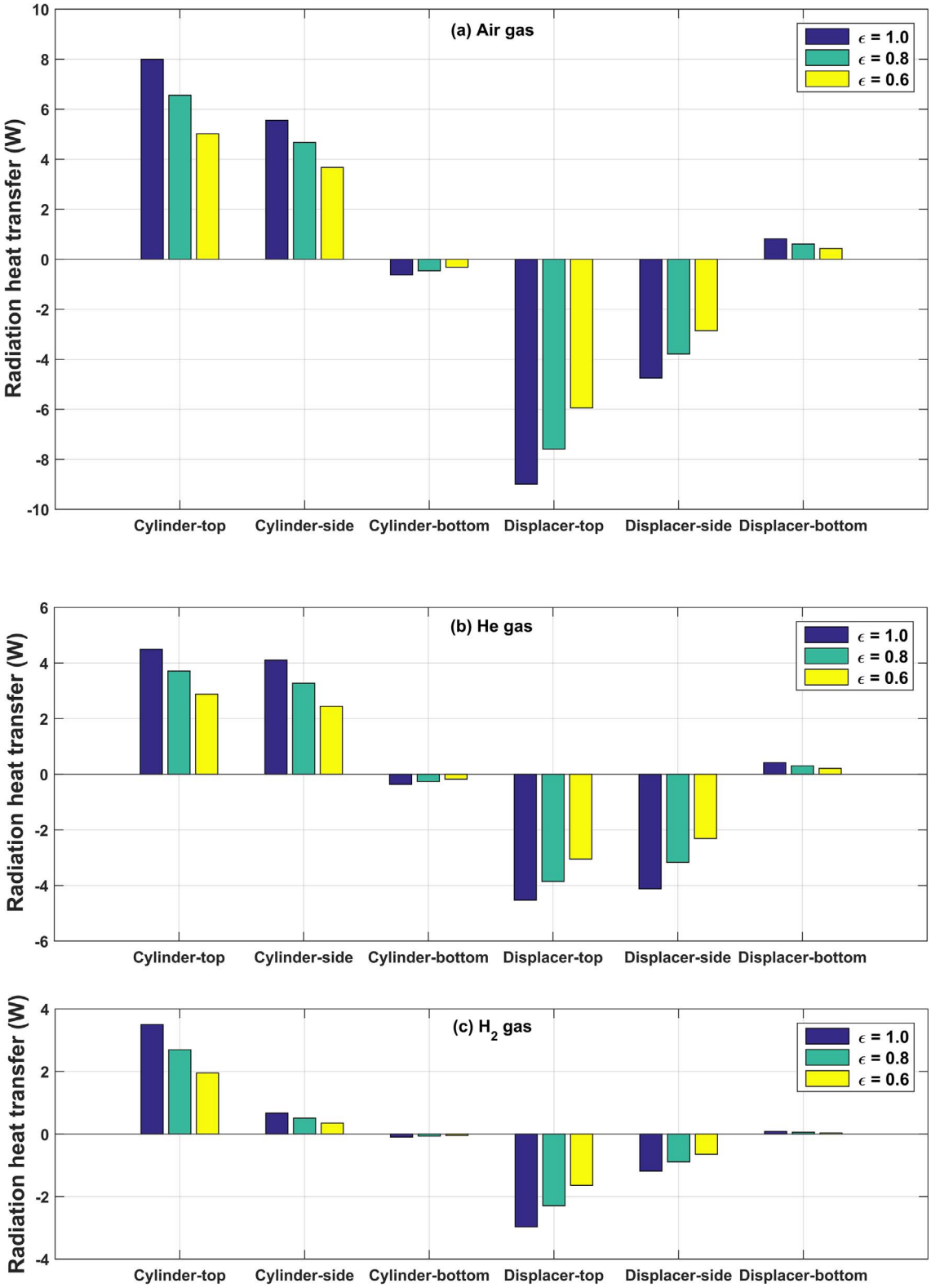


Fig. 12. Comparison of net average radiative heat transfer at each surface during one cycle for; (a) air gas, (b) He gas, and (c) H_2 gas.

model, the total amount of absorbed heat increases noticeably. For instance, in the case where air gas is considered as working fluid with an internal surface emissivity of the 0.6, the engine absorbs a total heat (Q_{in}) of 109.2 W. While only 100.4 W is recorded as absorbed heat (Q_{in}) when the radiative heat transfer is ignored as shown in Table 4. Additionally, when the surface emissivity increases to 0.8 and 1 the engine absorb 112.5 W and 116.0 W respectively. In fact, the heat transfer from the heating walls (cylinder-side and cylinder-top) to the internal part (displacer) by convective and radiative heat transfer. Therefore, as the emissivity value increases, the heating walls emit a large amount of heat through radiation to the internal parts, which demonstrate the high value of Q_{in} (Table 4) as the surface emissivity increases. More explanation can be shown in Fig. 4, which represents the temperature profile at the displacer. The displacer is set to be insulated in the

Table 6
Percentage of radiative energy of each face.

	ε	Cylinder-top	Cylinder-side	Cylinder-bottom	Displacer-top	Displacer-side	Displacer-bottom
Air	1	56%	39%	−4%	−63%	−33%	6%
	0.8	55%	39%	−4%	−64%	−32%	5%
	0.6	55%	40%	−3%	−65%	−31%	5%
He	1	50%	46%	−4%	−50%	−46%	5%
	0.8	51%	45%	−4%	−53%	−43%	4%
	0.6	52%	44%	−3%	−55%	−42%	4%
H2	1	82%	16%	−2%	−70%	−28%	2%
	0.8	83%	15%	−2%	−71%	−27%	2%
	0.6	84%	15%	−2%	−70%	−28%	2%

Table 7
Percentage of the total radiative energy to the total supplied energy.

	ε	Q_{rad}/Q_{in}
Air	1	12%
	0.8	11%
	0.6	8%
He	1	9%
	0.8	7%
	0.6	6%
H_2	1	2%
	0.8	2%
	0.6	1%

boundary condition. Thus, the total heat transfer to the displacer is zero. Nevertheless, its surface walls absorb radiation heat from the hot walls, especially in the narrow channel where the displacer-side is facing the cylinder-side directly. It is clear from Fig. 4 that the displacer walls become hotter in the upper part where the engine is heated. Also, the displacer temperature in the upper section (x > 0.13 m) increases significantly with the surface emissivity. For example, in the case of ε = 1, the upper part of the displacer reached a temperature of 600 K. Therefore, the cold gases coming from the cold zone will be heated by the cylinder-side wall as well as by the displacer-side wall simultaneously. Thus, the power output improves proportionally with the surface emissivity because the large emissivity value leads to high displacer temperature profile as depicted in Fig. 4. The radiative heat flux on the displacer is shown in Fig. 5. It is clear that as the emissivity value is reduced, the radiative heat flux is also reduced. There are two reasons for that. Firstly, some of the incident radiation is reflected as the displacer surface is not a blackbody (ε < 1). Secondly, at low emissivity value, the cylinder heating walls emit less amount of heat compared to the heat emitted by the black body (ε = 1). Furthermore, the lower part of the displacer (x < 0.095 m) shows positive radiative heat flux value which means it emits heat to the cold cylinder surface. However, this heat flux is small due to the small temperature differences between the cold surfaces in the lower part. On the other hand, the upper part of the displacer receives a high amount of heat flux (negative sign) because it is near to the upper part of the cylinder which has high-temperature magnitude. All of the absorbed radiative heat by the displacer is transferred to the gas through convection heat transfer. As a result, the power output is influenced proportionally with the emissivity as shown in Table 4. For instance, when the emissivity increases from 0.6 to 0.8 and 1, the power output rises from 8.8 W to 9.1 W and 9.6 W, respectively. All these power output values are greater than the power output when the radiation heat is ignored (7.9 W). Also, there is a slight improvement in thermal efficiency from 8.0% to 8.1 and 8.3% for emissivity values of 0.6, 0.8 and 1, respectively.

The temperature contours are shown in Fig. 6. It is shown that the gas enters the hot zone with temperature about 700 K near the cylinder-side wall, while the gas temperature near to the displacer-side wall leaves the narrow channel at a temperature around 600 K. The highest recorded temperature level is the case where the walls emissivity has a value of ε = 1. The temperature level decreases as surface emissivity

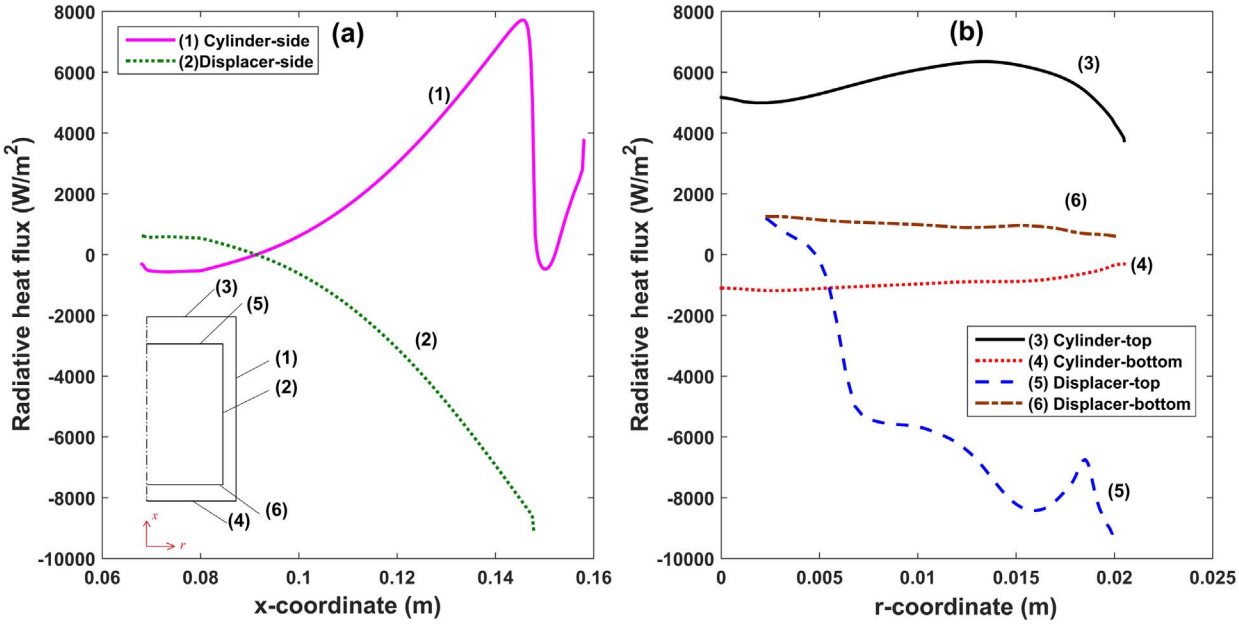


Fig. 13. Radiative heat flux at each face at the beginning of the cycle (air gas, ε = 1).

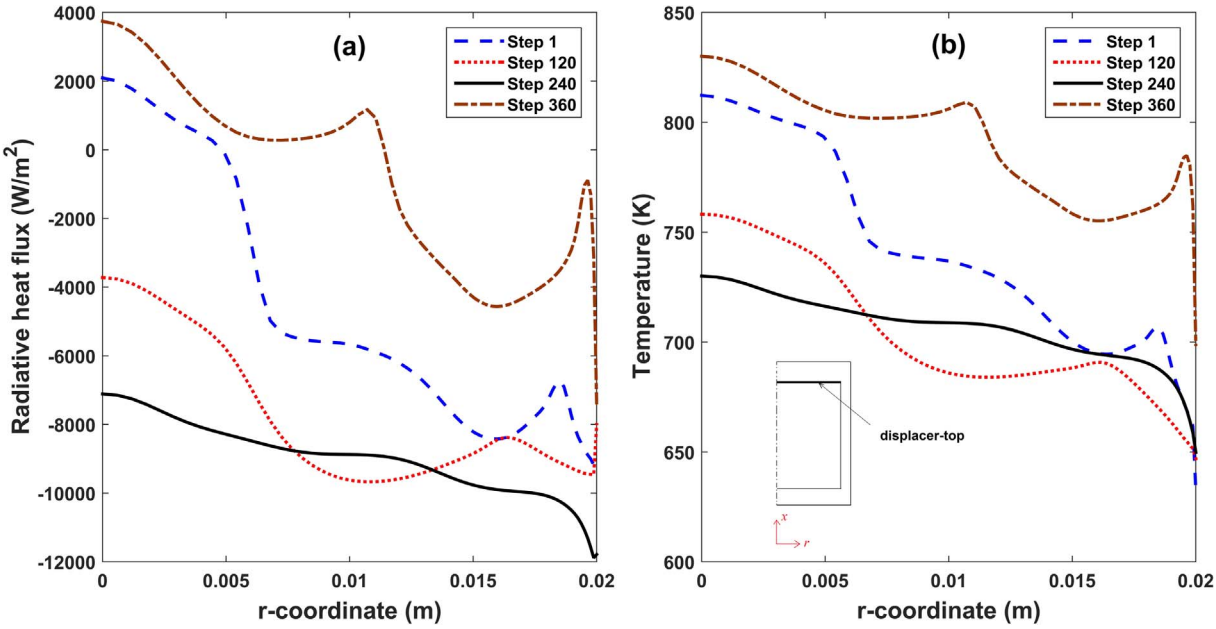


Fig. 14. Radiative heat flux and temperature profiles at the displacer-top surface at different time-steps of one cycle (480 steps) (air gas, $\varepsilon = 1$).

reduces. Also, the figure shows that the lowest temperature level belongs to the case where radiative heat transfer is ignored.

The air gas is then replaced by He gas in order to observe the impact of radiative heat transfer on the performance of Stirling engine with different gases. The He gas has very high thermal conductivity compared to that of air. It is 0.152 W/m·K at room temperature, while the thermal conductivity of air is only 0.0242 W/m·K. Fig. 7 depicts the radiative heat flux on the displacer for He gas as working fluid. By comparing He and air gases, the radiative heat flux is very small in the case of He gas, however, it changes proportionally with the emissivity value similar to the changing in the case of air gas. In the narrow channel, the He gas absorbs a high amount of heat from the cylinder-side wall through convection heat transfer (due to the high thermal conductivity of the He gas). Then, it is convected to displacer-side causing high-temperature profile in the displacer wall (Fig. 8). Thus, the temperature difference between the cylinder-side and the displacer-side surfaces becomes small. This explains the small values of radiative heat flux in the case of applying He gas (Fig. 7) compared to case of air gas (Fig. 5). Consequently, there is a small improvement in the power output with surface emissivity as expected. As the emissivity increases from 0.6 to 0.8 and 1, the power output rises respectively from 14.6 W to only 14.7 W and 14.8 W. While the thermal efficiency reduces slightly from 14.6% to 14.5% and 14.3%.

The H_2 gas is also examined in this work. It is known to have the highest thermal conductivity of all gases. Therefore, insignificant amount of radiative heat flux is recorded in the displacer-side surface (Fig. 9), due to the dominated convective heat transfer mode. In fact, due to the high thermal conductivity of H_2 gas, the surface of the displacer-side is heated by the gas to reach a high-temperature level. Hence, the displacer surface would be heated by the gas (by convection), and therefore, the radiation heat transfer could be ignored due to small temperature differences between the heating cylinder walls and the displacer surface (Fig. 10). Thus, the emissivity value has a trivial impact on radiative heat flux in the case of using H_2 gas as the working fluid, since the two surfaces have a close temperature level. The temperature contours for the three emissivity values (0.6, 0.8 and 1) are shown in Fig. 11. They are almost similar to the temperature contours of the case where radiation heat is ignored. Because the radiative heat flux is very small, the convective heat transfer is the dominated mode of heat transfer. Thus, the power output is not affected by the emissivity value. For all the emissivity values under study, the Stirling engine produces an almost same power output of 17.6 W. However, the output power is slightly greater than the case where radiation heat is ignored (17.4 W). The Stirling engine performance with He and H_2 gases are summarized in Table 5.

Fig. 12 depicts the net average radiative heat transfer at each face during one cycle. In the three gas type cases of (air, He, and H_2), the cylinder-top face emits the highest radiation energy. By considering the case of air gas with $\varepsilon = 1$ (Fig. 12(a)), the cylinder-top emits about 8 W of radiation energy, followed by the cylinder-side face (5.6 W). The cylinder-top surface is kept at a constant temperature of 800 K which is the highest temperature in the system, while, the surface temperature of cylinder-side is varying linearly from 800 K until 300 K (Eq. (1)). Therefore, as it is expected, the cylinder-top face has the highest net average emitted radiation heat transfer. Most of the emitted energy is absorbed by the displacer-top face (9 W). The cylinder-top and cylinder-side faces emit about 56% and 39% of the total emitted radiation energy, respectively. About 63% of the radiation energy is absorbed mainly by the displacer-top face, and 33% is absorbed by the displacer-side face. The remaining percentage of emitted and absorbed radiation energy occurs in the cold zone between the displacer-bottom and the cylinder-bottom faces.

When the He gas is used as the working fluid instead of air, the total net emitted radiation energy is reduced from 14.4 W to 9.0 W (for the case of $\varepsilon = 1$). However, the cylinder-top still emits the highest percentage of net average radiation heat energy (50%), flowed by the cylinder-side face (46%). The percentages of the average net radiation heat transfer for the other faces are summarized in Table 6.

On the other hand, when the H_2 gas is applied, the total net radiation energy to the total supplied energy does not exceed 2% as indicated in Table 7. Therefore, the convective heat transfer becomes dominant and the radiation heat transfer could be ignored. This is not like the cases of He and air gases where the percentage of radiative energy reaches up to 9% and 12% of the total supplied energy.

The data presented in Fig. 12 is the net average value of radiation energy for each face during one cycle. However, the radiative energy varies across the face and it varies at each time step during the cycle. Fig. 13 shows the profiles of the radiative heat flux of each face at the beginning of the cycle for the case of air gas with $\varepsilon = 1$. At the upper part of the cylinder-side ($x > 148$ mm), the cylinder-side face is no longer seeing the displacer-side face only. It sees also the surface of cylinder-top, displacer-top, and even it sees itself. At a distance of

x = 148 mm, the temperature of the cylinder-side is 736 K, while the cylinder-top is kept at a temperature of 800 K, and the displacer-top has a temperature greater than 736 K. Therefore, the cylinder-side receives a radiative energy from the cylinder-top and displacer-top faces. The things that demonstrates the sharp reduction in the radiative heat flux of the cylinder-side at a distance of x = 148 mm. As we move up in the x-direction, the temperature of the cylinder-side rises up and it becomes higher than the temperature of the displacer-top face. Accordingly, the cylinder-side starts to emit radiative energy to the displacer-top face, thus, the radiative heat flux profile rises up gradually after the sharp reduction point.

Furthermore, the radiative heat flux is varying widely at each surface, owing to the variation of the temperature profile at the each surface as represented in Fig. 14 for the surface of displacer-top.

6. Conclusions

In the current work, two radiation models (surface to surface model 'S2S' and Discrete Ordinates model 'DO') are examined to clarify the impact of radiation heat transfer on the performance of a β-type Stirling engine. The results revealed that ignoring radiation model leads to an error of 13% in the gained and rejected heat transfer. On the other hand, an error of 18% in power output and thermal efficiency is observed when air gas is used as the working fluid with a surface emissivity value of 1. The DO radiation model gives more accurate results, and it is less computational expense than the S2S radiation model. Furthermore, for air gas; high surface emissivity results in high power output and high thermal efficiency. This is attributed to the high contributed percentage of radiative heat transfer in the total supplied energy, which reaches up to 12% in the case of air gas with a surface emissivity value of 1. In contrast, for the H_2 gas, the convective heat transfer becomes dominating where it represents about 98% of the total supplied energy, and hence, the radiation energy could be ignored. Furthermore, in all case, the cylinder-top surface emits the highest radiation energy flowed by the cylinder-side surface. Most of this emitted radiation energy is absorbed by the displacer-top and the displacer-side.

Acknowledgment

The authors of this article sincerely acknowledge the support of King Fahd University of Petroleum & Minerals (KFUPM)–the Kingdom of Saudi Arabia, to this work.

References

[1] Shabanpour R, Sayyaadi H, Hosseinzade H. Sizing a solar dish Stirling micro-CHP system for residential application in diverse climatic conditions based on 3E analysis. Energy Convers Manage 2013;75:348–65.
[2] Boutammachte N, Knorr J. Field-test of a solar low delta-T Stirling engine. Sol Energy 2012;86:1849–56.
[3] Mahkamov K. An axisymmetric computational fluid dynamics approach to the analysis of the working process of a solar stirling engine. J Sol Energy Eng 2006;128:45–53.
[4] Reddy VS, Kaushik SC, Ranjan KR, Tyagi SK. State-of-the-art of solar thermal power plants — A review. Renew Sustain Energy Rev 2013;27:258–73.
[5] Ahmadi MH, Mohammadi AH, Dehghani S, Barranco-jiménez MA. Multi-objective thermodynamic-based optimization of output power of Solar Dish-Stirling engine by implementing an evolutionary algorithm. Energy Convers Manage 2013;75:438–45.
[6] Petrescu S, Petre C, Costea M, Malancioiu O, Boriaru N, Dobrovicescu A, et al. A methodology of computation, design and optimization of solar Stirling power plant using hydrogen / oxygen fuel cells. Energy 2010;35:729–39.
[7] Chan J, Wood JG, Schreiber JG. Development of advanced Stirling radioisotope generator for space exploration. National Aeronautics and Space Administration (Report No. E-15939); 2007.
[8] Chen NCJ, Griffin FP. A review of Stirling engine mathematical models. OAK Ridge National Laboratory (No. ORNL/CON-135); 1983 Aug.
[9] Dyson RW, Wilson SD, Tew RC. Review of computational Stirling analysis methods. National aeronautics and space administration. (No. AIAA–2004–5582); 2004.
[10] Gedeon D. Sage user 's guide. Sage v11 Edition, 2016. [Online]. Available: http://www.sageofathens.com/Documents/SageStlxHyperlinked.pdf. [accessed: 04-May-2016].
[11] Dyson RW, Wilson SD, Tew R, Demko R. Fast whole-engine Stirling analysis. National aeronautics and space administration (No. AIAA–2005–5558); 2005.
[12] Chen WL, Wong KL, Chang YF. A computational fluid dynamics study on the heat transfer characteristics of the working cycle of a low-temperature-differential γ-type Stirling engine. Int J Heat Mass Transf 2014;75:145–55.
[13] Salazar JL, Chen WL. A computational fluid dynamics study on the heat transfer characteristics of the working cycle of a β-type Stirling engine. Energy Convers Manage 2014;88:177–88.
[14] Abuelyamen A, Ben-Mansour R, Abualhamayel H, Mokheimer EMA. Parametric study on beta-type Stirling engine. Energy Convers Manage 2017;145:53–63.
[15] Makhkamov K, Ingham DB. Theoretical investigations on the Stirling engine working process. In: Energy conversion engineering conference and exhibit, 2000. (IECEC) 35th intersociety. vol. 1. IEEE; 2000. p. 101–10.
[16] Mahkamov K. Design improvements to a biomass stirling engine using mathematical analysis and 3D CFD modelling. Energy Resour Technol 2006;128:203–15.
[17] Lin C, Wang X, Chen X, Zhang Z. Improve the free-piston stirling engine design with high order analysis method. In: Applied mechanics and materials. Trans Tech Publications. vol. 44–47; 2011. p. 1991–5.
[18] Chen W, Wong K, Chang Y. A numerical study on the effects of moving regenerator to the performance of a β-type Stirling engine. Int J Heat Mass Transf 2015;83:499–508.
[19] Costa SC, Barreno I, Tutar M, Esnaola JA, Barrutia H. The thermal non-equilibrium porous media modelling for CFD study of woven wire matrix of a Stirling regenerator. Energy Convers Manage 2015;89:473–83.
[20] Costa SC, Barrutia H, Esnaola JA, Tutar M. Numerical study of the heat transfer in wound woven wire matrix of a Stirling regenerator. Energy Convers Manage 2014;79:255–64.
[21] Tanaka M, Yamashita I, Chisaka F. Flow and heat transfer characteristics of the stirling engine regenerator in an oscillating flow. JSME Int J 1990;33:283–9.
[22] Gedeon D, Wood JG. Oscillating-flow regenerator test rig: hardware and theory with derived correlations for screens and felts. NASA-Lewis Research Center (No. NASA CR-198442); 1996.
[23] Costa SC, Barrutia H, Ander Esnaola J, Tutar M. Numerical study of the pressure drop phenomena in wound woven wire matrix of a Stirling regenerator. Energy Convers Manage 2013; 67: 57–65.
[24] Costa SC, Tutar M, Barreno I, Esnaola JA, Barrutia H, García D, et al. Experimental and numerical flow investigation of Stirling engine regenerator. Energy 2014;72:800–12.
[25] Li Z, Haramura Y, Kato Y, Tang D. Analysis of a high performance model Stirling engine with compact porous-sheets heat exchangers. Energy 2014;64:31–43.
[26] Ibrahim M. 2-D CFD simulation of the heat transfer and fluid dynamics in an experimental model of the hot end of a Stirling engine. In: Energytech. IEEE. 2012. p. 1–11.
[27] Jiang N, Simon TW. Heat transfer and fluid dynamics measurements in the expansion space of a stirling cycle engine. In: International mechanical engineering congress and exposition. IMECE 2006–15631; 2006.
[28] Mabrouk MT, Kheiri A, Feidt M. Displacer gap losses in beta and gamma Stirling engines. Energy 2014;72:135–44.
[29] Chen W-L, Yang Y-C, Salazar JL. A CFD parametric study on the performance of a low-temperature-differential γ-type Stirling engine. Energy Convers Manage 2015;106:635–43.
[30] Hsieh YC, Hsu TC, Chiou JSÃ. Integration of a free-piston Stirling engine and a moving grate incinerator. Renew Energy 2008;33:48–54.
[31] Abdollahpour A, Ahmadi MH, Mohammadi AH. Thermodynamic model to study a solar collector for its application to Stirling engines. Energy Convers Manage 2014;79:666–73.
[32] Yaqi L, Yaling H, Weiwei W. Optimization of solar-powered Stirling heat engine with finite-time thermodynamics. Renew Energy 2011;36:421–7.
[33] Karmacharya S, Putrus G, Underwood CP, Mahkamov K, Mcdonald S, Alexakis A. Simulation of energy use in buildings with multiple micro generators. Appl Therm Eng 2014;62:581–92.
[34] Cardozo E, Erlich C, Malmquist A, Alejo L. Integration of a wood pellet burner and a Stirling engine to produce residential heat and power. Appl Therm Eng 2014;73:669–78.
[35] ANSYS fluent theory guide. Release 15.0. 2013.
[36] Xiao G, Chen C, Shi B, Cen K, Ni M. Experimental study on heat transfer of oscillating flow of a tubular Stirling engine heater. Int J Heat Mass Transf 2014;71:1–7.
[37] Bert J, Chrenko D, Sophy T, Le L. Simulation, experimental validation and kinematic optimization of a Stirling engine using air and helium. Energy 2014;78:701–12.
[38] Rogdakis ED, Antonakos GD, Koronaki IP. Thermodynamic analysis and experimental investigation of a Solo V161 Stirling cogeneration unit. Energy 2012;45:503–11.
[39] Aksoy F, Cinar C. Thermodynamic analysis of a beta-type Stirling engine with rhombic drive mechanism. Energy Convers Manage 2013;75:319–24.
[40] Karabulut H, Serdar H, Çınar C, Aksoy F. An experimental study on the development of a β-type Stirling engine for low and moderate temperature heat sources. Appl Energy 2009;86:68–73.